1. What is the standard alternating current frequency in India?

(A) 60 Hz
(B) 50 Hz
(C) 100 Hz
(D) 25 Hz

2. The time taken by an alternating quantity to complete one full cycle is called its:

(A) Frequency
(B) Time period
(C) Amplitude
(D) Phase angle

3. The relationship between frequency f and time period T is:

(A) f = 1 / T
(B) f = T^2
(C) f = 2 * pi * T
(D) f = T / 2

4. The angular frequency omega of a sinusoidal waveform is given by:

(A) 2 * pi * f
(B) pi * f
(C) f / (2 * pi)
(D) 2 * pi / f

5. For a pure sinusoidal wave of peak amplitude Vm, the RMS value is:

(A) Vm / sqrt(2)
(B) Vm * sqrt(2)
(C) 2 * Vm / pi
(D) Vm / 2

6. For a pure sinusoidal wave of peak amplitude Vm, the average value over a half-cycle is:

(A) 2 * Vm / pi
(B) Vm / pi
(C) Vm / sqrt(2)
(D) Vm / 2

7. The Form Factor of a pure sine wave is defined as RMS value / Average value, equal to:

(A) 1.11
(B) 1.414
(C) 0.707
(D) 1.57

8. The Peak Factor (Crest Factor) of a pure sine wave is defined as Peak / RMS, equal to:

(A) 1.414 (sqrt(2))
(B) 1.11
(C) 0.637
(D) 2.0

9. In a purely resistive AC circuit, the phase angle between voltage and current is:

(A) 0 degrees (in phase)
(B) 90 degrees lag
(C) 90 degrees lead
(D) 180 degrees

10. In a purely inductive AC circuit, current lags applied voltage by:

(A) 90 degrees
(B) 0 degrees
(C) 45 degrees
(D) 180 degrees

11. In a purely capacitive AC circuit, current leads applied voltage by:

(A) 90 degrees
(B) 0 degrees
(C) 45 degrees
(D) 180 degrees

12. The inductive reactance XL of an inductor L at frequency f is given by:

(A) 2 * pi * f * L
(B) 1 / (2 * pi * f * L)
(C) 2 * pi * f / L
(D) L / (2 * pi * f)

13. The capacitive reactance XC of a capacitor C at frequency f is given by:

(A) 1 / (2 * pi * f * C)
(B) 2 * pi * f * C
(C) 2 * pi * C / f
(D) C / (2 * pi * f)

14. What is the SI unit of capacitive and inductive reactance?

(A) Ohm
(B) Farad
(C) Henry
(D) Siemens

15. The average power consumed by a pure inductor or pure capacitor over a full cycle is:

(A) Zero
(B) Vm * Im
(C) 0.5 * Vm * Im
(D) Infinite

16. The power factor of an electrical circuit is defined as the cosine of phase angle (cos phi) or:

(A) R / Z
(B) Z / R
(C) X / R
(D) R / X

17. Real (Active) electrical power P in a single-phase AC circuit is expressed in:

(A) Watts (W)
(B) Volt-Amperes (VA)
(C) Volt-Amperes Reactive (VAR)
(D) Joules

18. Reactive power Q in an AC circuit is expressed in:

(A) Volt-Amperes Reactive (VAR)
(B) Watts
(C) Volt-Amperes
(D) Henry

19. Apparent power S in an AC circuit is expressed in:

(A) Volt-Amperes (VA)
(B) Watts
(C) VAR
(D) Ohm

20. The relationship connecting Apparent Power S, Real Power P, and Reactive Power Q is:

(A) S = sqrt(P^2 + Q^2)
(B) S = P + Q
(C) S = P * Q
(D) S = P - Q

21. In a series R-L-C circuit at resonance, the net reactance is:

(A) Zero
(B) Infinite
(C) Equal to R
(D) Purely capacitive

22. The resonant frequency f0 of a series R-L-C circuit is:

(A) 1 / [2 * pi * sqrt(L * C)]
(B) 2 * pi * sqrt(L * C)
(C) sqrt(L / C)
(D) 1 / (L * C)

23. At series resonance, the impedance of the circuit is:

(A) Minimum and equal to R
(B) Maximum
(C) Zero
(D) Infinite

24. At series resonance, the current in the circuit is:

(A) Maximum
(B) Minimum
(C) Zero
(D) Negative

25. The power factor of a series RLC circuit at resonance is:

(A) Unity (1.0)
(B) Zero lagging
(C) Zero leading
(D) 0.707

26. In a balanced star (Y) connected 3-phase system, the line voltage VL and phase voltage Vph satisfy:

(A) VL = sqrt(3) * Vph
(B) VL = Vph
(C) VL = Vph / sqrt(3)
(D) VL = 3 * Vph

27. In a balanced star (Y) connected 3-phase system, the line current IL and phase current Iph satisfy:

(A) IL = Iph
(B) IL = sqrt(3) * Iph
(C) IL = Iph / sqrt(3)
(D) IL = 3 * Iph

28. In a balanced delta connected 3-phase system, the line voltage VL and phase voltage Vph satisfy:

(A) VL = Vph
(B) VL = sqrt(3) * Vph
(C) VL = 3 * Vph
(D) VL = Vph / 3

29. In a balanced delta connected 3-phase system, the line current IL and phase current Iph satisfy:

(A) IL = sqrt(3) * Iph
(B) IL = Iph
(C) IL = Iph / sqrt(3)
(D) IL = 3 * Iph

30. Total active power in a balanced 3-phase load (star or delta) is given by:

(A) sqrt(3) * VL * IL * cos(phi)
(B) 3 * VL * IL * cos(phi)
(C) VL * IL * cos(phi)
(D) sqrt(3) * Vph * Iph

31. In a series R-L DC circuit, the time constant tau is given by:

(A) L / R
(B) R / L
(C) R * L
(D) 1 / (R * L)

32. In a series R-C DC circuit, the time constant tau is given by:

(A) R * C
(B) 1 / (R * C)
(C) C / R
(D) R / C

33. After one time constant tau, the current in an energizing series RL circuit reaches what percentage of maximum?

(A) 63.2%
(B) 36.8%
(C) 50.0%
(D) 99.3%

34. After one time constant tau, a discharging capacitor voltage falls to what percentage of initial value?

(A) 36.8%
(B) 63.2%
(C) 50.0%
(D) 0%

35. A transient state in an electrical circuit persists until:

(A) Stored energy reaches a new steady-state equilibrium
(B) Current becomes zero
(C) Voltage doubles
(D) Resistance heats up

36. In an uncharged capacitor, at the exact instant of switching (t = 0+), the capacitor behaves as a:

(A) Short circuit
(B) Open circuit
(C) Constant voltage source
(D) Inductor

37. In an unenergized inductor, at the exact instant of switching (t = 0+), the inductor behaves as an:

(A) Open circuit
(B) Short circuit
(C) Infinite capacitor
(D) Oscillator

38. The two-wattmeter method is used to measure power in a 3-phase:

(A) 3-wire balanced or unbalanced load
(B) Single-phase circuit
(C) 4-wire system only
(D) DC circuit

39. In the two-wattmeter method, the total 3-phase active power is:

(A) W1 + W2
(B) W1 - W2
(C) sqrt(3) * (W1 + W2)
(D) W1 * W2

40. In the two-wattmeter method, if both wattmeters show identical positive readings (W1 = W2), load power factor is:

(A) 1.0 (Unity)
(B) 0.5
(C) 0 (Zero)
(D) 0.866

41. In the two-wattmeter method, if one wattmeter reads zero (W1 = P, W2 = 0), the load power factor is:

(A) 0.5
(B) 1.0
(C) 0
(D) 0.866

42. In the two-wattmeter method, if one wattmeter reads negative (W1 = -W2), the load power factor is:

(A) 0 (Zero lag/lead)
(B) 0.5
(C) 1.0
(D) 0.707

43. The load power factor angle phi in the two-wattmeter method is calculated by tan(phi) = :

(A) sqrt(3) * (W1 - W2) / (W1 + W2)
(B) (W1 - W2) / (W1 + W2)
(C) sqrt(3) * (W1 + W2) / (W1 - W2)
(D) (W1 + W2) / sqrt(3)

44. A coil of resistance 6 Ohms and inductive reactance 8 Ohms has an impedance of:

(A) 10 Ohms
(B) 14 Ohms
(C) 2 Ohms
(D) 48 Ohms

45. What is the power factor of a coil with R = 6 Ohms and XL = 8 Ohms?

(A) 0.6 lagging
(B) 0.8 lagging
(C) 1.0
(D) 0.6 leading

46. The Quality factor Q of a series R-L-C resonant circuit is given by:

(A) (1 / R) * sqrt(L / C)
(B) R * sqrt(C / L)
(C) omega0 * R * C
(D) 1 / (omega0 * L)

47. Bandwidth (BW) of a resonant RLC circuit is related to resonant frequency f0 and Quality factor Q by:

(A) BW = f0 / Q
(B) BW = f0 * Q
(C) BW = Q / f0
(D) BW = sqrt(f0 * Q)

48. The half-power frequencies f1 and f2 of a series resonant circuit are frequencies where power drops to:

(A) 50% of maximum (current drops to 70.7%)
(B) 25% of maximum
(C) 10% of maximum
(D) Zero

49. In a parallel R-L-C circuit (ideal L and C in parallel with R), the resonant frequency is:

(A) 1 / [2 * pi * sqrt(L * C)]
(B) 2 * pi * sqrt(L * C)
(C) sqrt(L / C)
(D) Zero

50. In a parallel resonant circuit, the impedance at resonance is:

(A) Maximum (Dynamic resistance L / (C*R))
(B) Minimum
(C) Zero
(D) Equal to inductive reactance

51. A parallel resonant circuit is commonly referred to as an:

(A) Rejector circuit
(B) Acceptor circuit
(C) Attenuator
(D) Rectifier

52. A series resonant circuit is commonly referred to as an:

(A) Acceptor circuit
(B) Rejector circuit
(C) Limiter
(D) Integrator

53. The voltage magnification factor across capacitor C at series resonance equals:

(A) Q * V_supply
(B) V_supply / Q
(C) Q^2 * V_supply
(D) Zero

54. A voltage v(t) = 100 * sin(314*t + 30 deg) and current i(t) = 10 * sin(314*t - 30 deg) have a phase difference of:

(A) 60 degrees (current lagging)
(B) 0 degrees
(C) 30 degrees
(D) 90 degrees

55. For the above circuit (v=100V peak, i=10A peak, 60 deg lag), the active power P is:

(A) 2500 W
(B) 5000 W
(C) 1000 W
(D) 3535 W

56. Three identical impedances of (8 + j6) Ohms are star connected to a 400 V, 3-phase supply. Phase voltage is:

(A) 230.9 V
(B) 400 V
(C) 133.3 V
(D) 692.8 V

57. For the above star system (Vph = 230.9 V, Z = 10 Ohms), the phase current Iph is:

(A) 23.09 A
(B) 40.0 A
(C) 13.33 A
(D) 69.28 A

58. If the same three impedances (8 + j6 Ohms) are reconnected in delta across 400 V, the phase current is:

(A) 40.0 A
(B) 23.09 A
(C) 69.28 A
(D) 120.0 A

59. The ratio of power taken by a delta-connected load to that taken by the same load in star is:

(A) 3 : 1
(B) 1 : 3
(C) sqrt(3) : 1
(D) 1 : 1

60. In an RL circuit with DC excitation V, the current growth differential equation is:

(A) L * di/dt + R * i = V
(B) L * d2i/dt2 + R * i = V
(C) R * di/dt + L * i = V
(D) L * i + R * di/dt = 0

61. The complete analytical solution for current growth in a series RL circuit is:

(A) i(t) = (V/R) * [1 - exp(-R*t / L)]
(B) i(t) = (V/R) * exp(-R*t / L)
(C) i(t) = (V/R) * [1 + exp(-t/RC)]
(D) i(t) = (V/L) * t

62. In an RC circuit charging through resistor R, the capacitor voltage vc(t) is given by:

(A) vc(t) = V * [1 - exp(-t / RC)]
(B) vc(t) = V * exp(-t / RC)
(C) vc(t) = (V/R) * exp(-t / RC)
(D) vc(t) = V * [1 + exp(-t/RC)]

63. In an RC circuit discharging through resistor R, the capacitor voltage is:

(A) vc(t) = V0 * exp(-t / RC)
(B) vc(t) = V0 * [1 - exp(-t/RC)]
(C) vc(t) = V0
(D) vc(t) = 0

64. A series RLC circuit with R = 20 Ohms, L = 0.1 H, and C = 100 microfarads has a damping factor zeta equal to:

(A) (R / 2) * sqrt(C / L) = 0.316 (underdamped)
(B) 1.0 (critically damped)
(C) 2.5 (overdamped)
(D) 0 (undamped)

65. A second-order RLC circuit is critically damped when R equals:

(A) 2 * sqrt(L / C)
(B) sqrt(L / C)
(C) 0.5 * sqrt(L / C)
(D) 4 * L / C

66. The natural resonant frequency omega_n of an RLC circuit is:

(A) 1 / sqrt(L * C)
(B) sqrt(L / C)
(C) R / (2 * L)
(D) 1 / (R * C)

67. Damped natural angular frequency omega_d in an underdamped RLC circuit is:

(A) omega_n * sqrt(1 - zeta^2)
(B) omega_n * sqrt(zeta^2 - 1)
(C) omega_n / sqrt(1 - zeta^2)
(D) omega_n

68. The complex power S is mathematically defined in terms of phasor V and current phasor I as:

(A) S = V * I* (conjugate of I)
(B) S = V * I
(C) S = V* * I
(D) S = V / I

69. If reactive power Q is positive (+jQ), the load is:

(A) Inductive (lagging power factor)
(B) Capacitive (leading power factor)
(C) Purely resistive
(D) Non-linear

70. If reactive power Q is negative (-jQ), the load is:

(A) Capacitive (leading power factor)
(B) Inductive (lagging power factor)
(C) Purely resistive
(D) Superconducting

71. Capacitor banks are connected in parallel with industrial inductive loads primarily to:

(A) Improve power factor and reduce line current
(B) Increase active power consumption
(C) Increase harmonic distortion
(D) Step down voltage

72. Skin effect in AC conductors causes current density to be:

(A) Concentrated near the conductor surface
(B) Uniform throughout cross-section
(C) Concentrated at the geometric center
(D) Zero at the surface

73. Skin effect increases with an increase in:

(A) Frequency, conductor diameter, and permeability
(B) DC current
(C) Conductor resistivity
(D) Insulation thickness

74. Proximity effect in parallel AC conductors causes current redistribution due to:

(A) Mutual magnetic flux of adjacent conductors
(B) Electrostatic charge accumulation
(C) Thermal gradients
(D) Dielectric breakdown

75. The phase sequence of a 3-phase system RYB reversed is:

(A) RBY
(B) YRB
(C) BYR
(D) YBR

76. In an RLC series circuit excited by a step voltage V, the transient response is overdamped when:

(A) R > 2 * sqrt(L / C)
(B) R = 2 * sqrt(L / C)
(C) R < 2 * sqrt(L / C)
(D) R = 0

77. In an underdamped series RLC circuit, the damping ratio zeta is related to circuit parameters by:

(A) zeta = (R / 2) * sqrt(C / L)
(B) zeta = 2 * R * sqrt(L / C)
(C) zeta = 1 / (R * sqrt(L*C))
(D) zeta = R * L * C

78. The logarithmic decrement delta of an underdamped oscillating transient is defined as:

(A) ln(x1 / x2) = 2 * pi * zeta / sqrt(1 - zeta^2)
(B) x1 / x2
(C) zeta * omega_n
(D) 1 / zeta

79. In Laplace domain analysis of an inductor with initial current I0, the equivalent s-domain model is:

(A) Impedance s*L in series with opposing voltage source L*I0
(B) Impedance s*L only
(C) Impedance 1/(s*C)
(D) Resistance L

80. In Laplace domain analysis of a capacitor with initial voltage V0, the equivalent s-domain series model is:

(A) Impedance 1/(s*C) in series with voltage source V0 / s
(B) Impedance 1/(s*C) only
(C) Current source C*V0
(D) Resistance 1/C

81. A coil with resistance R and inductance L is switched onto AC voltage v(t) = Vm * sin(omega*t + alpha). Transient current is zero if switched at:

(A) alpha = theta = arctan(omega*L / R)
(B) alpha = 0
(C) alpha = 90 deg
(D) alpha = 180 deg

82. If the above coil is switched at alpha - theta = -pi/2, the initial DC offset current is:

(A) Maximum (equal to -Vm / |Z|)
(B) Zero
(C) Half of maximum
(D) Infinite

83. The maximum transient peak current under worst-case closing angle is termed the:

(A) Doubling effect (peak current approaches 2 * Vm / |Z|)
(B) Ferranti effect
(C) Skin effect
(D) Hall effect

84. In a parallel practical resonant tank (inductor with branch resistance R_L in parallel with ideal capacitor C), resonant frequency omega_r is:

(A) sqrt(1/(L*C) - (R_L^2 / L^2))
(B) 1 / sqrt(L * C)
(C) sqrt(1/(L*C) + (R_L^2 / L^2))
(D) R_L / L

85. For resonance to occur in the practical tank circuit above, which condition must hold?

(A) R_L < sqrt(L / C)
(B) R_L > sqrt(L / C)
(C) R_L = infinity
(D) L = C

86. Dynamic impedance Z_dyn of the practical parallel tank circuit at resonance is strictly real and equals:

(A) L / (C * R_L)
(B) R_L
(C) sqrt(L / C)
(D) C / (L * R_L)

87. In an unbalanced 3-phase 3-wire star load with isolated neutral, the neutral displacement voltage V_Nn is determined by:

(A) Millman's Theorem: Sum(V_k * Y_k) / Sum(Y_k)
(B) Thevenin theorem only
(C) Ohm's law
(D) Tellegen's theorem

88. Symmetrical components method (Fortescue's theorem) resolves an unbalanced 3-phase set of phasors into:

(A) Positive sequence, negative sequence, and zero sequence sets
(B) Two orthogonal sets
(C) Harmonic series
(D) DC components

89. The complex operator 'a' (alpha) in symmetrical components represents a phase rotation of:

(A) 120 degrees counterclockwise (exp(j * 2*pi / 3))
(B) 90 degrees
(C) 180 degrees
(D) 60 degrees

90. The relation 1 + a + a^2 equals:

(A) 0
(B) 1
(C) 3
(D) -1

91. In a balanced 3-phase system, the zero-sequence currents are identically:

(A) Zero
(B) Equal to line current
(C) Three times line current
(D) Indeterminate

92. In a 3-phase 3-wire delta system, zero sequence line currents cannot exist because:

(A) There is no neutral return path for zero sequence currents
(B) Line voltage is zero
(C) Impedance is infinite
(D) Phase angle is 90 deg

93. In the two-wattmeter method, when wattmeter readings are W1 = 1000 W and W2 = -500 W, the power factor is:

(A) cos(arctan(sqrt(3)*1500 / 500)) = cos(79.1 deg) = 0.189 lag
(B) 0.5 lag
(C) 0.866
(D) 0.707

94. Ferranti Effect in long transmission lines under light load conditions causes:

(A) Receiving end voltage to exceed sending end voltage
(B) Receiving end voltage to drop to zero
(C) Severe overheating
(D) Frequency shift

95. Ferranti effect is physically caused by the interaction of line inductance with:

(A) Line shunt charging capacitance
(B) Series resistance
(C) Skin effect
(D) Corona discharge

96. In AC power systems, Total Harmonic Distortion (THD) of a current waveform is defined as:

(A) sqrt(Sum_{n=2}^infinity I_n^2) / I_1
(B) I_1 / I_rms
(C) Sum(I_n) / I_1
(D) I_peak / I_rms

97. The crest factor of a square wave is:

(A) 1.0
(B) 1.414
(C) 1.11
(D) 2.0

98. The form factor of a symmetrical square wave is:

(A) 1.0
(B) 1.11
(C) 1.414
(D) 0.707

99. A pure triangular waveform has a form factor equal to:

(A) 2 / sqrt(3) = 1.155
(B) 1.11
(C) 1.414
(D) 1.0

100. In a non-sinusoidal AC system, the apparent power S is related to active P, reactive Q, and distortion power D by:

(A) S = sqrt(P^2 + Q^2 + D^2)
(B) S = P + Q + D
(C) S = sqrt(P^2 + Q^2) + D
(D) S = P * Q * D

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Derive the mathematical expressions for RMS value, Average value, Form Factor, and Crest Factor of a pure sinusoidal waveform.

Q2. Analyze the steady-state behavior of pure R, pure L, and pure C elements subjected to sinusoidal excitation, deriving impedance, phase, and power.

Q3. Derive the complete impedance, current, phasor diagram, and Power Triangle relationships for a series R-L-C AC circuit.

Q4. Derive the resonant frequency, quality factor, bandwidth, and voltage magnification of a Series R-L-C Resonant Circuit.

Q5. Analyze the Parallel Resonant Circuit (Tank Circuit with coil resistance), deriving resonant frequency and dynamic impedance.

Q6. Derive the mathematical relationships between Line and Phase quantities in a Balanced Star (Y) Connected 3-Phase System.

Q7. Derive the mathematical relationships between Line and Phase quantities in a Balanced Delta Connected 3-Phase System.

Q8. Derive the complete theory of the Two-Wattmeter Method for measuring 3-phase power and power factor in a 3-wire system.

Q9. Analyze the transient growth and decay of current in a series R-L circuit excited by a constant DC voltage source.

Q10. Analyze the transient charging and discharging of a capacitor in a series R-C circuit excited by a DC voltage source.

Q11. Formulate and solve the second-order differential equation for a series R-L-C circuit, analyzing overdamped, critically damped, and underdamped cases.

Q12. Explain the physical origin of Skin Effect and Proximity Effect in AC conductors, deriving formulas for skin depth.

Q13. Explain Fortescue's Symmetrical Components Method, defining symmetrical operators and transformation matrices for unbalanced 3-phase analysis.

Q14. Explain the Ferranti Effect in AC Transmission Lines, providing mathematical derivation and mitigation methods.

Q15. Derive the mathematical formula for power factor correction, determining the required rating of a shunt capacitor bank.

Q16. Analyze an AC circuit problem: Calculate total current, power, and power factor for a parallel R-L and R-C network.

Q17. Derive the transient current expression when an alternating voltage v(t) = Vm*sin(omega*t + alpha) is switched onto an R-L circuit.

Q18. Explain Harmonics in AC power systems, defining Total Harmonic Distortion (THD), Crest Factor, and Triplen harmonics in 3-phase circuits.

Q19. Explain the Unbalanced 3-Phase Star Load with Isolated Neutral, and derive the neutral shift voltage using Millman's Theorem.

Q20. Analyze the step response of an uncharged capacitor energized through an inductor and resistor (Series R-L-C step response).

Q21. Explain Complex Power in terms of Voltage and Current Phasors, defining Apparent, Active, Reactive, and Distortion Power.

Q22. Explain the Two-Wattmeter readings under different load power factors (Unity, 0.866, 0.5, 0, and leading power factor).

Q23. Explain Phase Sequence and phase sequence indicators (rotating disc type and static lamp type).

Q24. Derive the frequency response, cut-off frequencies, and bandwidth of a parallel resonant circuit with high Q.

Q25. Analyze an AC bridge: Maxwell's Inductance-Capacitance Bridge, deriving balance equations for L and R.

Q26. Analyze Hay's Bridge and explain why it is uniquely suited for measuring High-Q inductors.

Q27. Analyze Schering's Bridge for precision measurement of dielectric capacitance and loss angle (dissipation factor).

Q28. Explain Transient Analysis of an LC Tank circuit, deriving the equation for undamped harmonic oscillations.

Q29. Explain the phenomenon of Inrush Current in Power Transformers during AC energization.

Q30. Describe Polyphase System Symmetrical Components Power Invariance and the Fortescue Power Formula.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[B]	**Q21:**	[A]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[B]	**Q22:**	[A]	**Q42:**	[A]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[A]	**Q23:**	[A]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[A]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[A]	**Q25:**	[A]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[A]	**Q26:**	[A]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[A]	**Q27:**	[A]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[A]	**Q28:**	[A]	**Q48:**	[A]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[A]	**Q29:**	[A]	**Q49:**	[A]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[A]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[A]	**Q31:**	[A]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[A]	**Q32:**	[A]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[A]	**Q33:**	[A]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[A]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[A]	**Q35:**	[A]	**Q55:**	[A]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[A]	**Q36:**	[A]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[A]	**Q37:**	[A]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[A]	**Q38:**	[A]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[A]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[A]	**Q40:**	[A]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

- Reactance and Impedance: XC = 1 / (omega * C); Z_C = -j*XC = (1 / j*omega*C) = XC /_ -90 deg.
- Power: p(t) = (Vm*Im / 2) * sin(2*omega*t). Average Power: P_avg = 0 W.

>>Solution Question 8: Derive the resonant frequency, quality factor, bandwidth, and Power Transformation Series R-L-C Resonance Circuit.

applied supply voltage.
5. Bandwidth (BW) and Half-Power Frequencies:
- Half-power frequencies f1 and f2 occur where current drops to I_0 / sqrt(2) (power drops to P_max / 2):
omega_1,2 = - R/(2*L) +- sqrt([R/(2*L)]^2 + 1/(LC)).
- Bandwidth: Delta-omega = omega_2 - omega_1 = R / L => Delta-f = f_2 - f_1 = R / (2 * pi * L) = f_0 / Q.

>>Solution to Question 6: Derive the Parallel Resonant Circuit (Tank Circuit) with Phase Impedance in Balanced Star (Y) Connected 3-Phase System.

- Resolving into rectangular components:
V_RY = V_ph * [(1 + j0) - (-1/2 - j*sqrt(3)/2)] = V_ph * [3/2 + j*sqrt(3)/2].
- Magnitude:
|V_RY| = V_ph * sqrt((3/2)^2 + (sqrt(3)/2)^2) = V_ph * sqrt(9/4 + 3/4) = V_ph * sqrt(12/4)
= sqrt(3) * V_ph.
- Phase Angle:
theta = arctan([sqrt(3)/2] / [3/2]) = arctan(1 / sqrt(3)) = 30 degrees.
- General Law: In star connection, Line Voltage is sqrt(3) times Phase Voltage and leads the corresponding phase voltage by 30 degrees:
V_L = sqrt(3) * V_ph.
3. Current Relationship:
- Each line is connected directly in series with a single phase winding.
- Therefore, Line Current equals Phase Current: I_L = I_ph.
4. Total Active Power:
- P = 3 * V_ph * I_ph * cos(phi) = 3 * (V_L / sqrt(3)) * I_L * cos(phi) = sqrt(3) * V_L * I_L * cos(phi).

>> Solution to Question 8: Derive the mathematical relations for Two-Wattmeter Method for measuring 3-phase Balanced Delta-Connected 3-Phase System.

[Detailed Step-by-Step Answer]:
Theory of Two-Wattmeter Power Measurement:
1. Connection Scheme:
- Current coils (CC) of Wattmeters W1 and W2 are inserted in lines R and B respectively.
- Pressure coils (PC) are connected between lines R-Y and lines B-Y respectively (line Y serves as common reference).
2. Wattmeter Reading Equations:
- Wattmeter 1: Measures instantaneous product of line current I_R and voltage across its PC (V_RY).
Phasor V_RY leads V_RN by 30 deg. If phase current I_R lags V_RN by load angle phi, angle between V_RY and I_R is (30 deg - phi).
W1 = |V_RY| * |I_R| * cos(30 deg - phi) = V_L * I_L * cos(30 deg - phi).
- Wattmeter 2: Measures product of line current I_B and voltage across its PC (V_BY = -V_YB).
Angle between V_BY and I_B is (30 deg + phi).
W2 = |V_BY| * |I_B| * cos(30 deg + phi) = V_L * I_L * cos(30 deg + phi).
3. Sum of Readings (Total 3-Phase Power):
- W1 + W2 = V_L * I_L * [cos(30 - phi) + cos(30 + phi)].
Using identity cos(A - B) + cos(A + B) = 2 * cos(A) * cos(B):
W1 + W2 = V_L * I_L * [2 * cos(30 deg) * cos(phi)] = V_L * I_L * [2 * (sqrt(3)/2) * cos(phi)] = sqrt(3) * V_L * I_L * cos(phi) = Total Active Power P.
4. Difference of Readings and Power Factor Determination:
- W1 - W2 = V_L * I_L * [cos(30 - phi) - cos(30 + phi)] = V_L * I_L * [2 * sin(30 deg) * sin(phi)] = V_L * I_L * sin(phi).
- Dividing difference by sum:
(W1 - W2) / (W1 + W2) = [V_L * I_L * sin(phi)] / [sqrt(3) * V_L * I_L * cos(phi)] = tan(phi) / sqrt(3).
- Thus: tan(phi) = sqrt(3) * (W1 - W2) / (W1 + W2).
- Load Power Factor: cos(phi) = cos(arctan[sqrt(3) * (W1 - W2) / (W1 + W2)]).

Solution to Question 9:

>>Solution Step 1: Write the differential equation for a series R-L-C circuit excited by a DC voltage.

Response: i(t) = exp(-alpha*t) * [A1 * cos(omega_d*t) + A2 * sin(omega_d*t)]. Ringing, damped sinusoidal oscillation.

>>Solution Question 12 Solution Question 13 Solution Question 14 Solution Question 15 Solution Question 16 Solution Question 17 Solution Question 18 Solution Question 19 Solution Question 20 Solution Question 21 Solution Question 22 Solution Question 23 Solution Question 24 Solution Question 25 Solution Question 26 Solution Question 27 Solution Question 28 Solution Question 29 Solution Question 30 Solution Question 31 Solution Question 32 Solution Question 33 Solution Question 34 Solution Question 35 Solution Question 36 Solution Question 37 Solution Question 38 Solution Question 39 Solution Question 40 Solution Question 41 Solution Question 42 Solution Question 43 Solution Question 44 Solution Question 45 Solution Question 46 Solution Question 47 Solution Question 48 Solution Question 49 Solution Question 50 Solution Question 51 Solution Question 52 Solution Question 53 Solution Question 54 Solution Question 55 Solution Question 56 Solution Question 57 Solution Question 58 Solution Question 59 Solution Question 60 Solution Question 61 Solution Question 62 Solution Question 63 Solution Question 64 Solution Question 65 Solution Question 66 Solution Question 67 Solution Question 68 Solution Question 69 Solution Question 70 Solution Question 71 Solution Question 72 Solution Question 73 Solution Question 74 Solution Question 75 Solution Question 76 Solution Question 77 Solution Question 78 Solution Question 79 Solution Question 80 Solution Question 81 Solution Question 82 Solution Question 83 Solution Question 84 Solution Question 85 Solution Question 86 Solution Question 87 Solution Question 88 Solution Question 89 Solution Question 90 Solution Question 91 Solution Question 92 Solution Question 93 Solution Question 94 Solution Question 95 Solution Question 96 Solution Question 97 Solution Question 98 Solution Question 99 Solution Question 100

- Va = Va0 + Va1 + Va2
- Vb = Va0 + a^2 * Va1 + a * Va2
- Vc = Va0 + a * Va1 + a^2 * Va2

4. Analysis Matrix (Inverse Transformation):

- [Va0, Va1, Va2]^T = (1/3) * [1 1 1] [Va]
[1 a a^2] [Vb]
[1 a^2 a] [Vc].

>>Solution to Question Bank of the Fundamental Form of a Transmission Line Equation: Derivation and Solution Methods

- Initial load state:
Active Power: P.
Initial Power Factor: cos(phi1) => Initial Reactive Power Q1 = P * tan(phi1).
- Target compensated state:
Target Power Factor: cos(phi2) => New Net Reactive Power Q2 = P * tan(phi2).
- Required Reactive Power supplied by Shunt Capacitor Bank:
Q_C = Q1 - Q2 = P * [tan(phi1) - tan(phi2)].
3. Capacitor Sizing (Single-Phase):
- Reactive power generated by capacitor across voltage V at frequency omega = 2*pi*f:
Q_C = V^2 / XC = V^2 * (omega * C).
- Equating expressions:
C = Q_C / (2 * pi * f * V^2) = P * [tan(phi1) - tan(phi2)] / [2 * pi * f * V^2].
4. 3-Phase Bank Configuration:
- For a Delta-connected capacitor bank across line voltage V_L:
C_delta = Q_C / [3 * 2 * pi * f * V_L^2].
- Delta connection requires 1/3 the capacitance of a Star connection for identical kVAR rating because each unit operates at full line voltage V_L rather than phase voltage V_L / sqrt(3).

>>Solution to Question 14: Analyze the AC circuit problem. Calculate the load power factor for a parallel RL load. Switch to an R-L circuit.

AC Transient Analysis of Series R-L Circuit:
1. Differential Equation:
- L * di/dt + R * i(t) = Vm * sin(omega * t + alpha), with switch closed at t = 0 (i(0) = 0).
2. General Solution (Complementary Function + Particular Integral):
- Natural / Transient Response (CF):
i_tr(t) = K * exp(- R*t / L).
- Steady-State Response (PI):
i_ss(t) = (Vm / |Z|) * sin(omega * t + alpha - theta),
where |Z| = sqrt(R^2 + omega^2*L^2) and impedance angle theta = arctan(omega*L / R).
- Total current:
i(t) = K * exp(- R*t / L) + (Vm / |Z|) * sin(omega * t + alpha - theta).
3. Evaluation of Constant K from Initial Condition i(0) = 0:
- 0 = K + (Vm / |Z|) * sin(alpha - theta) => K = - (Vm / |Z|) * sin(alpha - theta).
- Complete Analytical Solution:
i(t) = (Vm / |Z|) * [sin(omega*t + alpha - theta) - sin(alpha - theta) * exp(- R*t / L)].
4. Physical Significance:
- Case A (Zero Transient): If circuit is switched at angle alpha such that alpha - theta = 0 or pi, sin(alpha - theta) = 0. The DC transient term vanishes identically; current enters steady state immediately!
- Case B (Maximum Doubling Transient): If switched when alpha - theta = -pi/2, sin(alpha - theta) = -1. The DC offset is maximum: K = +Vm / |Z|. After a half-cycle, peak current approaches nearly 2 * (Vm / |Z|).

>> Solution to Question 18: Explain Harmonics in AC power systems, defining Total Harmonic Distortion (THD), Crest Factor, and Triplen harmonics in 3-phase circ

[Problem Statement]: Explain the Unbalanced 3-Phase Star Load with Isolated Neutral, and derive the neutral shift voltage using Millman's Theor
[Detailed Step-by-Step Answer]:
Unbalanced Star Load with Isolated Neutral (Neutral Shift):
1. Problem Nature:
- An unbalanced star load consisting of unequal phase admittances Y_R, Y_Y, Y_B is connected to a balanced 3-phase supply (voltages V_RN, V_YN, V_BN relative to supply neutral N).
- The load neutral point 'n' is isolated (floating, no neutral wire).
- Because impedances are unequal, the load neutral 'n' shifts away from supply neutral 'N', establishing a non-zero Neutral Displacement Voltage V_nN.
2. Derivation of V_nN via Millman's Theorem:
- Applying KCL at isolated load neutral 'n':
I_Rn + I_Yn + I_Bn = 0.
- Express currents in terms of load branch voltages:
(V_R - V_n) * Y_R + (V_Y - V_n) * Y_Y + (V_B - V_n) * Y_B = 0.
- Expand terms:
V_R * Y_R + V_Y * Y_Y + V_B * Y_B = V_n * (Y_R + Y_Y + Y_B).
- Since supply neutral N is 0 V (V_R = V_RN, etc.):
V_nN = [V_RN * Y_R + V_YN * Y_Y + V_BN * Y_B] / [Y_R + Y_Y + Y_B].
3. Physical Hazard:
- Once V_nN is non-zero, the voltages across the individual load phases become unequal:
V_load_R = V_RN - V_nN.
- A phase with high impedance can experience severe overvoltage, destroying connected equipment.

>> [Problem Statement]: Analyze the step response of an uncharged capacitor energized through an inductor and resistor (Series R-L-C step response).
Solution Question 20

>>Solution to Question 22 Explain the Term Power in terms of Voltage and Current. Power factor. Active, Reactive and Distortion Power.

W1 = - W2; Total active power P = W1 + W2 = 0.

>>Solution to Question 28: Phase Sequence and Resonance: Series resonance occurs when the inductive and capacitive reactances are equal in magnitude, with high Q.

- Lower cut-off frequency: omega_1 = omega_0 - omega_0 / (2*Q).
- Upper cut-off frequency: omega_2 = omega_0 + omega_0 / (2*Q).

4. Bandwidth:

- BW = omega_2 - omega_1 = omega_0 / Q => f_2 - f_1 = f_0 / Q.
- Higher Q results in narrower bandwidth and superior frequency selectivity.

>>Solution to Question 26 Analyze the AC Bridge Maxwell's Inductance Capacitance Bridge, deriving balance conditions for L and R.

Lx = (C1 * R2 * R3) / (1 + omega^2 * C1^2 * R1^2).

4. High-Q Simplification:

- Coil Quality factor Q = omega * Lx / Rx = 1 / (omega * C1 * R1).
- For Q > 10: (omega * C1 * R1)^2 = (1/Q)^2 < 0.01 << 1.
- The term (1 + omega^2 * C1^2 * R1^2) approximates strictly to 1.
- Simplified formula:

Lx approx C1 * R2 * R3.

- Extremely simple and highly accurate for high-Q reactors.

>> Solution Question 28: Derive Schering Bridge for AC Tank circuit and derive the equation for unknown loss angle (dissipation factor).

where omega_0 = 1 / sqrt(L * C).
3. General Harmonic Solution:
- vc(t) = A * cos(omega_0 * t) + B * sin(omega_0 * t).
- Applying initial conditions:
vc(0) = A = V0.
i(0) = - C * dvc/dt |_{t=0} = - C * [- A * omega_0 * sin(0) + B * omega_0 * cos(0)] = - C * B * omega_0 = 0 => B = 0.
- Final Expressions:
vc(t) = V0 * cos(omega_0 * t).
i(t) = V0 * sqrt(C / L) * sin(omega_0 * t) = I_peak * sin(omega_0 * t).
4. Physical Energy Oscillation:
- Electric energy in capacitor: E_E(t) = 0.5 * C * [vc(t)]^2 = 0.5 * C * V0^2 * cos^2(omega_0 * t).
- Magnetic energy in inductor: E_M(t) = 0.5 * L * [i(t)]^2 = 0.5 * L * [V0^2 * (C/L)] * sin^2(omega_0 * t) = 0.5 * C * V0^2 * sin^2(omega_0 * t).
- Total Energy: E_total = E_E(t) + E_M(t) = 0.5 * C * V0^2 * [cos^2 + sin^2] = 0.5 * C * V0^2 = Constant.
- Continuous, undamped sloshing of energy between electric and magnetic domains at natural frequency f0.

>> Solution to Question 20: Describe the Power System Components Power Transformer during AC Generation.

[Detailed Step-by-Step Answer]:
Power Invariance in Symmetrical Components:
1. Total Instantaneous 3-Phase Power in Phase Coordinates:
- Complex power: S_3ph = Va * Ia* + Vb * Ib* + Vc * Ic* = [V_abc]^T * [I_abc]*.
2. Substitution of Symmetrical Component Transformation:
- Let [A] be the Fortescue symmetrical component transformation matrix:
[V_abc] = [A] * [V_012], and [I_abc] = [A] * [I_012].
- Substitute:
S_3ph = ([A] * [V_012])^T * ([A] * [I_012])* = [V_012]^T * [A]^T * [A]* * [I_012]*.
3. Evaluation of [A]^T * [A]*:
- [A] = [1 1 1]
[1 a^2 a]
[1 a a^2].
- Evaluating product:
[A]^T * [A]* = 3 * [I] (where [I] is 3x3 identity matrix).
4. Power Equation:
- S_3ph = 3 * [V_012]^T * [I_012]* = 3 * (Va0 * Ia0* + Va1 * Ia1* + Va2 * Ia2*).
- Active Power: P_3ph = 3 * [Va0*Ia0*cos(phi0) + Va1*Ia1*cos(phi1) + Va2*Ia2*cos(phi2)].
- Reactive Power: Q_3ph = 3 * [Va0*Ia0*sin(phi0) + Va1*Ia1*sin(phi1) + Va2*Ia2*sin(phi2)].
- Power Invariance Factor: The factor of 3 arises because standard Fortescue transformation is non-unitary; dividing [A] by sqrt(3) yields power-invariant unitary matrices.
